

Data Analytics Capstone

An Investigative Agentic AI Framework for CMS Hospital Excess Readmission Risk: Parallel
Machine Learning, Fairness Monitoring, and Tool-Grounded Quality Improvement Case
Dossiers under Human-in-the-Loop Review

Synopsis

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QM640: Data Analytics Capstone

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July 19, 2026

Introduction

Background and Context

Hospital readmissions remain a central quality and cost concern in U.S. healthcare operations. Under the Centers for Medicare & Medicaid Services (CMS) Hospital Readmissions Reduction Program (HRRP), hospitals with higher-than-expected unplanned 30-day readmissions face Medicare payment reductions of up to three percent (Centers for Medicare & Medicaid Services [CMS], 2024). Hospital quality leaders therefore need early risk insight, equitable monitoring across reporting cycles, and actionable quality-improvement (QI) options that are faithful to public CMS evidence.

Supervised machine learning (ML) fits tabular CMS quality files for excess-risk scoring, and TRIPOD+AI emphasizes transparent validation and fairness reporting for prediction models (Collins et al., 2024). Audits show readmission models can encode contextual bias (Wang, Weiner, Saria, & Kharrazi, 2024), and fairness can drift over time after deployment (Davis et al., 2025). These findings motivate parallel statistical association, predictive scoring, and fairness-monitoring tracks—but they do not by themselves produce a board-ready QI case package.

The harder operational gap is investigative decision support. When Excess Readmission Ratio (ERR) exceeds 1.0, quality teams must (a) assemble the hospital’s CMS metrics, (b) find peer hospitals that improved across fiscal releases, (c) compare what changed, (d) run simple what-if calculations (for example, how much predicted readmission rate would need to fall for $ERR \leq 1$), and (e) propose QI options with citations. That multi-step casework is slow when done manually. Unconstrained large language models (LLMs) are fast but unsafe: they hallucinate rates, invent peers, and recommend unsupported interventions (Singhal et al., 2023). Tool-wielding agents can explore clinical tables (Yang et al., 2026) and LLMs can narrate SHAP

outputs in ICU settings, yet prior work has not evaluated a multi-step investigative agent that builds peer-benchmarked, what-if-supported, citation-verified QI case dossiers on multi-year CMS HRRP public files under mandatory human-in-the-loop (HITL) review.

This capstone therefore runs two parallel product tracks on the same verified CMS dataset (FY2024–FY2026; pooled eligible $N = 35,724$ hospital–condition rows with non-missing ERR). Track A delivers association, ML scoring, and fairness monitoring. Track B delivers an Investigative Agentic QI Case Agent whose value is case investigation and dossier assembly—not ML output captioning and not autonomous CMS approval. Research questions are independent so timeline risk is controlled.

Problem Statement

U.S. acute-care hospitals face avoidable Medicare payment exposure under HRRP when excess 30-day readmission risk ($ERR > 1.0$) is not anticipated, when subgroup performance is not monitored across CMS fiscal releases, and when quality leaders lack a trustworthy investigative dossier that correctly combines hospital metrics, improving peers, temporal change, and what-if analysis before human QI decisions. Prior CMS HRRP studies mostly stop at descriptive analytics or ML prediction; GenAI readmission work mostly uses clinical notes or unconstrained chat. Using public CMS Provider Data Catalog HRRP and Hospital General Information files for FY2024–FY2026, this study addresses complementary problems in parallel: (1) associate hospital attributes with excess risk; (2) score excess-risk status with supervised ML out-of-time; (3) monitor fairness/performance across fiscal releases; and (4)—the novel Agentic/GenAI problem—evaluate a multi-step tool-grounded investigative agent that opens high-ERR cases, discovers improving peers, computes what-if sensitivities, and produces a verified QI case dossier for mandatory human Accept/Edit/Reject. Dependent variable for

scoring $Y = 1$ if $ERR > 1.0$ (else 0); independent variables X include ownership, hospital type, emergency services, state/ZIP proxies, measure/condition, discharge volume, and related public attributes. Agentic success is defined by higher evidence completeness, lower numeric hallucination, and higher HITL usefulness than (a) a no-tool LLM and (b) a static dashboard export—not by auto-approving CMS policy.

Purpose of the Study

The purpose is to deliver a Q1-aligned, timeline-safe package that (a) explains excess-risk associations, (b) validates an ML scoring engine, (c) monitors fairness across FY2024–FY2026, and (d) evaluates an Investigative Agentic QI Case Agent that uniquely solves multi-source case investigation and dossier construction under HITL governance. The GenAI contribution is evidence that tool-grounded multi-step investigation beats both unconstrained LLM narration and static scorecards for faithfulness and actionable QI case support.

What the Agent Uniquely Solves (Value Chain)

The agent is not an ML narrator. It is a QI case investigator. Table A1 states the unique value chain.

Step	Agent work (tool-grounded)	Why this is not simple ML interpretation	HITL gate
1. Case intake	Open high-ERR hospital–condition cases from CMS FY files	Operational triage across thousands of rows	Human sets priority rules
2. Evidence gather	Pull hospital metrics + measure definitions via tools	Multi-source retrieval, not a SHAP caption	Logged tool JSON only
3. Peer learning	Find peers	Comparative discovery	Human reviews peer list

	(ownership/state/condition) whose ERR improved FY24→FY26	ML scoring alone does not perform	
4. Temporal diagnosis	Quantify what changed (ERR, predicted/expected rates, volume)	Cross-year investigation workflow	Human validates relevance
5. What-if planning	Calculator tool: change needed in predicted rate for $ERR \leq 1$	Counterfactual planning beyond prediction scores	Human judges feasibility
6. QI options	Propose 2–3 candidate QI levers with citations to tool evidence	Decision options package, not a probability	Accept / Edit / Reject
7. Verifier pass	Second agent/tool check removes unsupported numeric claims	Anti-hallucination control layer	Safety metric reported
8. Dossier delivery	Board-ready case dossier labeled ADVISORY ONLY	End-to-end case management value chain	Mandatory human review

Policy boundary. The agent never auto-approves CMS penalties, clinical orders, payment adjustments, or model retraining/deployment. Every dossier is advisory; accountability remains with the human quality leader.

Technology Responsibility Matrix

Table A clarifies parallel tracks. Guiding rule: ML scores; fairness analytics monitors; tools compute facts; the agent investigates and assembles dossiers; humans decide.

Layer	What it solves / does	What it does NOT do	Primary RQ
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Statistics	Associate hospital attributes with $ERR > 1.0$	Build QI dossiers; approve actions	RQ1
Traditional ML + SHAP	Score excess-risk probability out-of-time; optional driver ranks as one evidence source	Replace investigative peer/what-if casework; auto-approve decisions	RQ2
Fairness / temporal monitoring	Measure AUROC and ΔFNR across FY releases and subgroups	Draft QI options; invent peers	RQ3
Deterministic tools	get_case, find_improving_peers, temporal_delta, whatif_err, measure_definition, optional shap, audit_log	Write free-text strategy without evidence; approve QI	Supports all
Investigative Agentic GenAI (planner + verifier)	Run multi-step case investigation and produce citation-verified QI dossiers	Interpret ML only; auto CMS/model policy decisions	RQ4
Human-in-the-loop governance	Accept / Edit / Reject every dossier before operational use	Delegating accountability to AI	RQ4 + policy

Problem / Solution / Achievement. Problem: excess-risk insight, fairness monitoring, and trustworthy QI casework without autonomous CMS decisions. Solution: parallel stats/ML/fairness tracks plus an investigative agentic dossier pipeline. Achievement: publishable

scoring/fairness results and evidence that the agent improves dossier faithfulness, evidence completeness, and HITL usefulness versus no-tool LLM and static dashboard baselines.

Scope and Objectives

Scope boundaries. U.S. Medicare-participating acute-care hospitals in CMS HRRP joined to same-year Hospital General Information for FY2024–FY2026 (measures: AMI, HF, PN, COPD, CABG, HIP-KNEE). Track B agent produces advisory QI case dossiers only. No PHI, no Kaggle, no auto clinical/CMS actions. Only core HRRP + General Information files (plus optional enrichment) are in analytic scope.

Primary objectives (parallel). (1) Association analysis for excess risk. (2) Out-of-time ML scoring. (3) Fairness/performance monitoring across FY releases. (4) Evaluate Investigative Agent QI Case Agent (peer discovery + temporal change + what-if + verified dossier) against no-tool LLM and static dashboard baselines under mandatory HITL. (5) Reproducible GitHub package with tool logs and dossier score sheets.

Independence design (timeline safety). RQ1–RQ4 are parallel. RQ1 remains valuable if ML gains are small. RQ2 reports absolute AUROC/calibration even if Δ AUROC is modest. RQ3 is monitoring and does not require RQ1/RQ2 Ha. RQ4 can use CMS tools alone; ML/SHAP is optional evidence inside the dossier, so RQ4 is not blocked if RQ2 underperforms.

Research Question 1 (RQ1)

Which hospital structural and contextual variables are significantly associated with excess readmission risk ($ERR > 1.0$) after adjusting for condition measure and fiscal-year release?

Hypothesis RQ1. H0: After adjustment, focal predictors have no association ($\beta = 0$). Ha: At least one focal predictor has a nonzero association at $\alpha = .05$.

Research Question 2 (RQ2)

How well can supervised ML classifiers (regularized logistic regression, random forest, gradient boosting) discriminate excess readmission risk on held-out and out-of-time (later FY) samples relative to a transparent logistic baseline?

Hypothesis RQ2. H0: Best ML out-of-time AUROC $\leq$ logistic baseline + 0.02. Ha: Best ML out-of-time AUROC $>$ logistic baseline + 0.02. Absolute AUROC, PR-AUC, and calibration are standalone deliverables regardless of Ha.

Research Question 3 (RQ3)

Do discrimination and fairness metrics (e.g., AUROC and Δ FNR across ownership/rurality subgroups) differ across FY2024, FY2025, and FY2026 when models trained on earlier releases are scored on later releases?

Hypothesis RQ3. H0: Fairness gaps and AUROC do not differ across releases beyond sampling variation. Ha: At least one operationally meaningful change ($|\Delta$ FNR| increase $>$ 0.05 or AUROC drop $>$ 0.03).

Research Question 4 (RQ4) — Investigative Agentic / GenAI use case

For high-ERR hospital-condition cases, does a multi-step tool-grounded Investigative Agentic QI Case Agent (case intake $\rightarrow$ metrics $\rightarrow$ improving-peer discovery $\rightarrow$ temporal change analysis $\rightarrow$ what-if calculator $\rightarrow$ citation-linked dossier $\rightarrow$ verifier) produce higher evidence completeness, lower numeric hallucination, and higher HITL Accept/Edit usefulness than (a) a no-tool LLM and (b) a static dashboard export alone, when every dossier is advisory only under mandatory human review?

Hypothesis RQ4. H0: The investigative agent does not improve faithfulness, evidence completeness, or HITL usefulness versus baselines. Ha: The agent improves at least two of

{faithfulness, evidence completeness, HITL usefulness} at $\alpha = .05$. Policy constraint: no autonomous CMS, clinical, or model-deployment actions.

Sample Size Calculation

Sample size is planned per RQ; tabular N uses the maximum of RQ1–RQ3 minima. RQ4 uses a separate paired case sample.

RQ1. Green's rule $n \geq 50 + 8m$ with $m \approx 15 \rightarrow n \geq 170$; conservative target ≥ 400 .

RQ2. AUROC ≈ 0.75 , half-width $\approx 0.03 \rightarrow N_{\text{RQ2}} \approx 800$.

RQ3. Medium correlation $\rho = 0.30$ for gap vs time $\rightarrow \approx 84$ stratum–period units.

RQ4. Paired dossier comparison (agent vs baselines) on the same cases. For $dz \approx 0.35$ – 0.50 at $\alpha = .05$, power = .80, $n \approx 34$ – 67 ; planning value $N_{\text{RQ4}} \geq 80$ high-ERR cases stratified by condition and ownership. Independent of RQ2 Ha.

Sample Size Summary Table

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1	Power / Green's rule	$\alpha = .05$; Power = .80; $m \approx 15$	170 (Green); 400 (conservative)
RQ2	CI precision for AUROC	AUROC ≈ 0.75 ; half-width ≈ 0.03	800 hospital–condition records
RQ3	Power for correlation	$\alpha = .05$; Power = .80; $\rho = 0.30$	84 stratum–period units
RQ4	Paired power (dossier metrics)	$\alpha = .05$; Power = .80; $dz \approx 0.35$ – 0.50	≥ 80 high-ERR paired cases

Final sample size decision. Tabular modeling uses pooled eligible $N = 35,724$. Out-of-time ML trains on earlier FY and tests on later FY. Agentic evaluation uses ≥ 80 high-ERR investigative cases with paired dossiers across systems.

Data Description

Dataset verification status (July 13–19, 2026). Verified CMS public sources: DATASET/hospitals_annual_2024.zip, DATASET/hospitals_annual_2025.zip, and DATASET/theme_hospitals_current/ (FY2026). Not Kaggle. Core extracts in DATASET/analytic_core/ with dataset_metadata.json. These data support Track A (scoring/fairness) and Track B (investigative agent tools for peers, temporal deltas, and what-if inputs).

Files Used for This Capstone (In Scope)

Only the following files are used for RQ1–RQ4. Other theme CSVs remain archived/out of scope.

Fiscal Year	HRRP File (analytic_core)	General Information File	Source Archive / Snapshot	Rows	ERR Non-Missing (Eligible N)	Performance Period
FY2024	FY2024_HRRP_Hospital.csv	FY2024_Hospital_General_Information.csv	hospitals_annual_2024.zip → hospitals_2024-10-30.zip	18,774	12,077	07/01/2019–06/30/2022
FY2025	FY2025_HRRP_Hospital.csv	FY2025_Hospital_General_Information.csv	hospitals_annual_2025.zip	18,500	11,920	07/01/2022–06/30/2025

25	pital.csv	ormation.csv	025.zip → hospitals_2025-11- 26.zip	10	7	20– 06/30/20 23
FY20 26	FY2026_HRRP_Hos pital.csv	FY2026_Hospital_General_Inf ormation.csv	theme_hospitals_c urrent	18,3 30	11,72 0	07/01/20 21– 06/30/20 24

Pooled metadata. Total HRRP rows = 55,614; eligible ERR rows = 35,724; unique facilities $\approx$ 2,946. Six measures: READM-30-AMI/HF/PN/COPD/CABG/HIP-KNEE-HRRP. Join key Facility ID. Dictionary: HOSPITAL_Data_Dictionary.pdf.

Optional enrichment: Unplanned_Hospital_Visits-Hospital.csv, HCAHPS, selected HVBP/spending. Out of scope: ASC*, REH*, PCH*, maternal, imaging, unused quarterly duplicates.

Primary URLs and paths:

1. <https://data.cms.gov/provider-data/topics/hospitals>
2. <https://data.cms.gov/provider-data/dataset/9n3s-kdb3>
3. Local: DATASET/hospitals_annual_2024.zip; hospitals_annual_2025.zip;
theme_hospitals_current/
4. Local analytic: DATASET/analytic_core/ (+ dataset_metadata.json)
5. <https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatient-pps/hospital-readmissions-reduction-program-hrrp>
6. GitHub (to publish): <https://github.com/<your-username>/qm640-investigative-qi-agent-hrrp>

Ethics and privacy. Public hospital-level de-identified metrics; no PHI. Agent uses approved tools only and must cite tool JSON. Limitations: ecological inference; proxy fairness

attributes; overlapping CMS windows; residual LLM risk mitigated by verifier + mandatory HITL.

Data Dictionary (Mandatory)

Official CMS definitions are in HOSPITAL_Data_Dictionary.pdf. Table 1 = modeling fields; Table 2 = investigative agent case-log fields.

Variable (CMS column)	Type	Role	Definition / Units	RQ Link
Facility ID	ID	Key	CMS hospital identifier	All
Facility Name	Text	Descriptor	Hospital name	Context
State	Categorical	Predictor / peer filter	U.S. state	RQ1, RQ3, RQ4
Hospital Ownership	Categorical	Predictor / fairness / peer filter	From General Information	RQ1–RQ4
Hospital Type	Categorical	Predictor / filter	Acute Care Hospitals primary	RQ1, RQ2
Emergency Services	Binary/Text	Predictor	Emergency services indicator	RQ1, RQ2
ZIP Code	Categorical	Fairness / geography proxy	Urban/rural derivation	RQ3, RQ4
Measure Name	Categorical	Stratum	Six HRRP condition measures	All
Number of Discharges	Numeric	Predictor / what-if context	Eligible discharges	RQ1, RQ2, RQ4
Predicted Readmission Rate	Numeric (%)	Context / what-if input	Hospital predicted rate	RQ1, RQ4

Expected Readmission Rate	Numeric (%)	Context / what-if input	Expected rate given mix	RQ1, RQ4
Excess Readmission Ratio	Numeric	Continuous risk / case trigger	Predicted / expected	RQ1, RQ4
excess_risk_flag (derived)	Binary	Dependent Y / case intake	1 if ERR > 1.0	RQ2–RQ4
Start Date / End Date	Date	Time index	Performance window	RQ3, RQ4
fiscal_year_label (derived)	Ordinal	Time cohort	FY2024 / FY2025 / FY2026	RQ3, RQ4
peer_improve_flag (derived)	Binary	Agent evidence	1 if peer ERR decreased across FYs	RQ4
whatif_delta_predicted (derived)	Numeric	Agent evidence	Change in predicted rate needed for ERR ≤ 1	RQ4

Agent Case-Log Field	Type	Role	Definition	RQ Link
case_id	ID	Key	Hospital–condition–FY investigative case	RQ4
tools_called	JSON	Audit	Ordered tools: case, peers, temporal, what-if, definitions, optional SHAP	RQ4
system_arm	Categorical	Comparator	no_tool_llm static_dashboard	RQ4

			investigative_agent	
evidence_completeness	Numeric 0–1	Primary metric	Required evidence slots filled with tool-backed facts	RQ4
faithfulness_score	Numeric 0–1	Primary metric	Claims entailed by tool outputs / factual claims	RQ4
numeric_hallucination_rate	Numeric 0–1	Safety metric	Incorrect numbers / numbers stated	RQ4
hitl_decision	Categorical	Governance	accept edit reject	RQ4
hitl_usefulness	Ordinal 0–4	Secondary metric	Blind-scored dossier usefulness	RQ4
advisory_only	Binary	Policy	1 = labeled advisory; never auto-applied	RQ4

Inclusion: non-missing ERR in FY2024–FY2026, joinable General Information when available. Exclusion: suppressed ERR. Unit: hospital–condition–fiscal-year. Eligible N = 35,724. RQ4 samples $\geq$ 80 high-ERR cases for dossier evaluation.

Analytic Approach

Architecture (parallel tracks; see Tables A and A1). Track A — Stats/ML/Fairness: clean CMS tables → EDA → association models → out-of-time classifiers → SHAP → fairness/temporal metrics. Track B — Investigative Agentic QI Case Agent: case intake → tool investigation (metrics, improving peers, temporal deltas, what-if) → planner dossier → verifier → HITL Accept/Edit/Reject. ML/SHAP may feed Track B as optional evidence but is not

required for agent evaluation. The agent does not choose model retraining or CMS policy actions.

Solution to RQ1

Multivariable logistic regression (elastic-net optional). Report adjusted ORs, 95% CIs, VIF. $\alpha = .05$.

Solution to RQ2

Compare regularized logistic, random forest, and gradient boosting. Cross-validate on earlier FY; lock test on later FY.

Metrics: AUROC; Accuracy = $(TP+TN)/(TP+TN+FP+FN)$; Precision = $TP/(TP+FP)$; Recall = $TP/(TP+FN)$; F1; Brier; ECE. SHAP optional for Track B evidence slots.

Solution to RQ3

Train on earlier FY; score later FY without refit. Compute AUROC, Brier, ΔFNR by ownership/rurality (Wang et al., 2024; Davis et al., 2025). Meaningful change thresholds: $|\Delta FNR| > 0.05$ or AUROC drop > 0.03 .

Solution to RQ4 (Investigative Agentic QI Case Agent)

What the agent uniquely solves. Multi-step QI case investigation and dossier assembly that currently consumes analyst time and that unconstrained LLMs corrupt with hallucinations. This is a value chain of intake $\rightarrow$ peer learning $\rightarrow$ temporal diagnosis $\rightarrow$ what-if $\rightarrow$ options $\rightarrow$ verification $\rightarrow$ HITL—not post-hoc ML interpretation.

Tools (deterministic). `open_high_err_case();` `get_hospital_metrics();`
`find_improving_peers();` `compute_temporal_deltas();` `whatif_err_to_target();`
`get_measure_definition();` `get_shap_optional();` `write_case_audit();`

Agent loop. Planner (ReAct-style) calls tools, fills a fixed dossier schema (Risk snapshot; Peer improvers; What changed; What-if; Candidate QI levers; Citations; Uncertainty). Verifier rejects claims not entailed by tool JSON. Banner: ADVISORY ONLY — HUMAN REVIEW REQUIRED.

Comparators. (B0) No-tool LLM. (B1) Static dashboard export (tables/plots only, no agent). (A1) Investigative agent + verifier (proposed).

Metrics. Evidence completeness; faithfulness; numeric hallucination rate; HITL usefulness (0–4); Accept/Edit/Reject rates. Paired tests on the same cases. Success: A1 beats B0 and B1 on ≥ 2 primary metrics without increasing reject rate due to unsafe content.

Recommendation and Application

Target users. Primary: hospital quality directors/committees needing investigative QI case dossiers for HRRP exposure. Secondary: analytics and AI-governance officers assessing GenAI safety under HITL.

Intended application. Deploy Track A for scoring/monitoring dashboards and Track B for on-demand investigative dossiers on high-ERR cases. Humans retain all operational decisions. Value versus status quo: faster than manual peer/temporal investigation; safer than ChatGPT; richer than a static scorecard.

Ten-week feasibility. Weeks 1–2: EDA/analytic_core. Week 3: RQ1. Weeks 4–5: RQ2. Week 6: RQ3. Weeks 7–8: RQ4 investigative agent tools, verifier, paired dossier evaluation. Weeks 9–10: packaging/presentation. Parallel RQs protect completion.

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Appendix A: Mentor Alignment Notes

This revision addresses mentor feedback end-to-end: (1) RQ1–RQ4 are parallel/timeline-safe; (2) Agentic GenAI is repositioned from post-ML interpretation or auto maintenance approval to an Investigative QI Case Agent value chain (peer discovery, temporal diagnosis, what-if, verified dossier, mandatory HITL). Replace student/mentor/GitHub placeholders before Olympus upload.